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MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT RANGE -5 TO +10g UNITS, INTEGRALLY LIGHTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers integrally lighted -5 to +10g range, aircraft accelerometers.

1.2 Classification. Accelerometers covered by this specification shall be of the following types, as specified (see 6.2):

ABU-4A/A - Nonhermetically sealed
ABU-4B/A - Hermetically sealed

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-L-6085	Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-P-26514	Polyurethane Foam, Rigid and Flexible, for Packaging
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover

FSC 6610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS23009	Accelerometer-Aircraft, Integrally Lighted, ABU-4A/A and ABU-4B/A
MS28105	Window, Dial-Aircraft Instrument Cover, Glass
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instruments

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3.1 Preproduction. This specification makes provisions for preproduction testing (see 4.4 and 6.2).

3.2 Materials

3.2.1 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to occur in storage or normal service.

3.2.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.2 Nonmagnetic materials. Nonmagnetic materials shall be used, except where magnetic materials are essential.

3.2.3 Lubricants. When required, the lubricant used shall be in accordance with MIL-L-6085. The amount of lubricant used shall be sufficient for at least 1,000 hours of operation.

3.2.4 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of

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any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Design and construction

3.4.1 Design. The accelerometer shall be designed to indicate aircraft accelerations in maneuvers and rough air, and shall conform to MS23009 and figure 1.

3.4.2 Response. The accelerometer shall be designed to comply with the response characteristics specified in 4.6.4.

3.4.3 Construction. The construction shall be reasonably simple to permit overhaul or repair with the use of tools and fixtures normally available commercially. The accelerometer shall be so constructed as to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure.

3.4.4 Pivots, bearings, and gears shall neither bind nor shake and shall be as near frictionless as practicable.

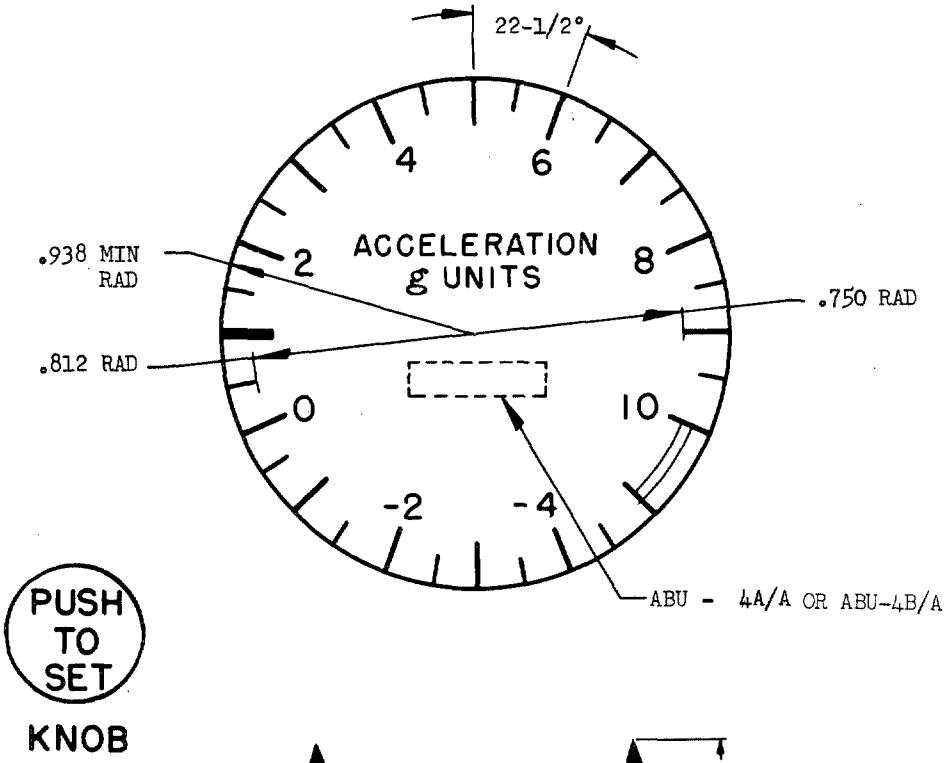
3.4.5 Case. The case shall conform to MS23009. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, Color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein.

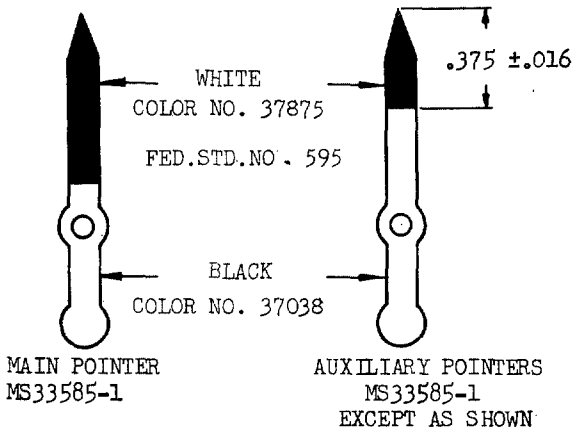
3.4.5.1 Hermetic sealing (ABU-4B/A). The case shall be hermetically sealed and shall be so designed that the internal mechanism may be removed, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. Hermetic sealing shall be such that the sealing will not be affected by the action of any atmosphere to which the indicator may be subjected. The case shall meet the necessary pressure requirements when tested for leaks by means of a mass spectrometer-type of helium leak detector.

3.4.5.2 Filling medium (ABU-4B/A). The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.4.5.3 Filling medium tube. The filling medium tube for the hermetically sealed unit shall be located on the rear of the case and shall not interfere with the locking knob operation or the electrical connection.

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DECIMALS ±.010
ANGLES ± 1/2°

FIGURE 1. Dial and pointers.

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3.4.6 Cover glass. The cover glass shall conform to MS28105 and MIL-G-81704.

3.4.6.1 Reflection-reducing coating. All cover glass and lighting wedges shall be provided with a reflection reducing coating that meets the requirements of MIL-C-14806. This requirement can be accomplished by obtaining a certificate of compliance from the manufacturer of the anti-reflective coating that states the coating meets the requirements of MIL-C-14806.

3.4.7 Dial. The dial shall conform to figure 1 and shall be fastened securely to the case or upon the frame of the mechanism by not less than two screws in such manner that it will not loosen or turn when the accelerometer is vibrated.

3.4.7.1 Distance from dial to cover glass. The distance between dial and cover glass shall be not greater than 0.250 inch.

3.4.8 Pointers. The accelerometer shall have three pointers; one main pointer for indicating instantaneous acceleration, and two auxiliary pointers, one for indicating maximum positive acceleration, and one for indicating maximum negative acceleration. The pointers shall conform to figure 1. They shall be light and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.4.9 Ratchet mechanism. Each auxiliary pointer shall be provided with a ratchet device, in order that indication of the maximum plus and minus deflections of the main pointer will be maintained until the auxiliary pointers are reset to obtain a new reading.

3.4.9.1 Auxiliary pointer return mechanism. A mechanism shall be provided for resetting the auxiliary pointers into juxtaposition with the main pointer. A knob, located as shown on MS23009, shall be provided for operating the return mechanism. The shaft of this knob shall be capable of withstanding a load of 15 ±1 pound applied perpendicular to the shaft at the extreme outward end of the knob without permanent deformation. Stops shall be provided in order that the auxiliary pointers, when reset, shall not indicate a lower or higher reading than that indicated by the main pointer. Stops shall be a part of the mechanism connecting the pointers and shall not directly engage them. Operation of the return mechanism shall require 3 to 8 pounds of force on the knob.

3.4.9.2 Main pointer mechanism. The main pointer mechanism shall not engage the mechanism of the auxiliary pointers when the imposed acceleration is less than the acceleration indicated by the auxiliary pointers.

3.4.10 Stops for acceleration-responsive element. Mechanical stops shall be provided for arresting the motion of the acceleration-responsive element and the pointers when the accelerometer is subjected to accelerations beyond the limits specified. One stop shall engage the element at an acceleration of not less than -5.1g and not more than -5.5g, and the other at an acceleration of not less than +10.1g and not more than +10.5g.

3.4.11 Locking device. A device for manually locking the position of the acceleration-responsive element shall be provided in order to protect the mechanism of the accelerometer from possible damage during shipment. The device shall be controlled by means of a suitable knurled knob, located at the rear of the case and marked as shown on MS23009. The force required to lock and unlock the unit shall be 4.5 ± 1 pound.

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3.4.12 Adjustment. The mechanism shall be provided with means to adjust or correct the indications. This means of adjustment shall be reasonably simple to manipulate and of a type that adjustment can be accomplished with tools ordinarily possessed by an instrument repairman.

3.4.13 Indication and range. The accelerometer shall indicate vertical accelerations when the plane of the mounting lugs is vertical and the +5.0 graduation is at top center. When in this position, normal gravity shall be indicated at +1g. The accelerometer shall be designed to indicate accelerations in the range -5 to +10g.

3.4.14 Integral lighting. The indicator shall incorporate two separate parallel integral lighting circuits. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. In the event of conflict between this specification and either MIL-L-25467 or MIL-L-27160, this specification shall govern. Lighting shall operate on 5V ac or dc. Lighting acceptability of production indicators shall be based on visual comparison with a prime standard. The method of lighting shall be approved by the procuring activity prior to fabrication.

3.4.14.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas covering the entire face of the indicator shall be measured in foot-lamberts for brightness. A minimum of 10 areas covering the entire face of the indicator shall be measured for color. The prime standard instrument and its recorded overall brightness and color measurements shall be approved by the procuring activity. The standard shall be checked at least every 50 hours of operation or 120 days, whichever occurs first, to determine any change in brightness or color. Variations greater than ±20 percent in two or more areas shall require return of the standard for corrective action. The data and, if required by the procuring activity, the prime standard shall be submitted for evaluation and reapproval or corrective action, or both.

3.4.15 Reliability

3.4.15.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.4.15.2 Reliability in mean time between failure (MTBF). The accelerometer shall have a minimum of 1,000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.7.

3.4.15.3 Longevity. The accelerometer shall have a total operating life of 3,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.6 Performance. The accelerometer shall be capable of passing the tests specified in Section 4 when subjected to the following conditions:

a. Temperature ranging from +71°C to -65°C.

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b. Relative humidity up to 100 percent.

c. Absolute pressures ranging from 30 inches mercury (Hg) down to 1.31 inches Hg approximately 70,000 feet altitude.

d. Fungus growth as encountered in tropical climates.

e. Exposure to salt sea atmosphere for a period of 50 hours.

f. Temperature shock in accordance with MIL-STD-810.

g. Acceleration and shock forces as anticipated.

h. When vibrated at a double amplitude (see 6.4.2) of 0.018 to 0.020 inch and at a frequency variation between 500 and 3000 cycles per minute (cpm) for a total of 3 hours, the main pointer shall not exceed 0.2g to 0.6g range and the auxiliary pointers shall not change more than 0.2g.

i. Static scale error shall not exceed 0.2g when subjected to the test specified in 4.6.10.

j. Static scale error and friction shall not exceed ±0.2g or as specified when tested in accordance with 4.6.2.

k. Dynamic scale error shall not exceed ±0.2g at any specified test point.

l. With the accelerometer revolved about a magnetic compass with a magnetic field intensity of 0.17 to 0.19 oersted, the compass shall not deflect more than 1 degree.

m. The mounting lugs shall withstand a load of 175 pounds without fracture.

n. Response characteristics shall be within the limits specified in 4.6.4, when the accelerometer is vibrated at frequencies from 5 to 20 Hz at maximum g forces for each test point.

3.7 Markings. All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters except the letter g, placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.7.1 The following markings shall be finished with lusterless white material, Color No. 37875 of FED-STD-595. The dimensions of the markings, when applicable shall be as specified in table I.

3.7.2 Durable lusterless black. The markings ABU-4A/A or ABU-4B/A, the background of the dial, and all other markings not otherwise specified, shall be finished in durable lusterless black, Color No. 37038 of FED-STD-595. The height of the markings shall be 0.062 ±0.016 inch.

3.8 Weight of each accelerometer shall not exceed 1.50 pounds.

3.9 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

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TABLE I. Dimensions and markings.

Markings	Height or length (inch ±0.016)	Width of line or graduation (inch ±0.005)
Numerals	0.188	MS33558
Lettering "g Units" (6.4.3) and "Acceleration"	0.062	MS33558
Major graduations (except +1 "g")	See figure 1	0.031
+1 "g" graduations	See figure 1	0.062
Minor graduations	See figure 1	0.015
Arc-section dial	---	0.015
Knob markings	0.062	MS33558

3.10 Finishes and protective coatings

3.10.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows.

3.10.1.1 Dials, small holes, and case inserts need not be anodized.

3.10.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.10.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.10.1.4 Castings, containing non-aluminum-alloy integral inserts, may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.10.1.5 When abrasion resistance is a factor, chemical films conforming to MIL-C-5541 shall not be used in lieu of anodizing.

3.10.2 Steel parts. Steel parts shall be cadmium-plated, where practicable, in accordance with type II or III, as applicable, of QQ-P-416, and of a class that is adequate to achieve the degree of protection required.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Envelope. An envelope, furnished by the contractor, shall contain and be packaged with each instrument and shall be marked as follows:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTALLATION INSTRUCTIONS,
MOUNTING SCREWS, AND REMOVABLE SPRING NUTS"

3.13 Workmanship. The accelerometer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner.

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Particular attention shall be given to neatness, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.13.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practice. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the accelerometer, they shall be held or limited accordingly.

3.13.2 Screw assemblies. Assembly screws shall be tight. The word, tight, means that they cannot be appreciably tightened further without damage or injury to the screw or threads.

3.13.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.13.4 Soldering. Soldering shall be performed in accordance with MIL-STD-454.

3.13.5 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.13.6 Cleaning. The accelerometer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection and testing of the accelerometer shall be classified as follows:

a. Preproduction tests . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg and at room temperature, approximately 25°C). Unless otherwise specified, the accelerometer shall be tested without correction for temperature or atmospheric pressure variations.

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4.3.2 Tapping. Unless otherwise specified, the accelerometer shall be tested in normal operating position and shall be lightly tapped (see 6.4.1) before a test reading is taken. Tapping shall be applied in the axis perpendicular to the dial at 60 Hz with a maximum double amplitude of 0.002 inch.

4.3.3 Vibration stand. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3,000 cpm, and shall subject the accelerometer to such vibration that a point on the case will describe, in a plane inclined 45 degrees to the horizontal plane, a circle the diameter of which is equal to the amplitude specified herein.

4.3.4 Calibration procedure. The contractor shall furnish a drawing of the mechanism and case showing clearly the vertical distance of the center of mass of the actuating mechanism from the center of the instrument dial for each of the required test points. This drawing shall also include a table of the revolutions per minute (rpm) required for each test point computed on the basis of a rotating arm whose radius to the center of the instrument dial is 20 inches. The rpm required for each test point shall be computed by the formula:

$$N = 9.5493 \sqrt{\frac{32.2g}{r}}$$

Where: N = rpm of rotating arm
g = test point
r = radius of arm in feet, corrected for mass position

4.3.4.1 Test points. N shall be computed for all test points from -5g to +10g in increments of 1g.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of nine accelerometers representative of the production equipment and shall be tested in a place and manner designated in the contract, purchase order, or invitation for bids (see 6.2). Three of the samples shall be subjected to and shall meet all the tests specified under 4.6. The remaining six accelerometers shall be subjected to and shall meet the qualification reliability and longevity requirements of 4.7.1 and 4.7.3.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.2).

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6, and the qualification phase of 4.7.

4.5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests . . . . . See 4.5.1

b. Sampling plan and tests . . . . . See 4.5.2

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4.5.1 Individual tests. Each accelerometer shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Static scale error and friction . . . . . See 4.6.2

c. Dynamic scale error. . . . . See 4.6.2.1

d. Individual test of MIL-L-25467 and MIL-L-27160 . . . . See 4.6.3

e. Return mechanism operation . . . . . See 4.6.5

f. Locking mechanism operation. . . . . See 4.6.5.1

g. Leakage . . . . . See 4.6.20

h. Fogging . . . . . See 4.6.21

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Three accelerometers shall be selected at random from each 200 or less produced on the contract or order and subjected to the following as described under 4.5.1 and 4.6.

a. Individual tests . . . . . See 4.5.1

b. Response characteristics . . . . . See 4.6.4

c. Force on knob . . . . . See 6.5.2

d. Low-temperature operation . . . . . See 4.6.6

e. High-temperature operation . . . . . See 4.6.7

f. Magnetic effect . . . . . See 4.6.8

g. Vibration error . . . . . See 4.6.9

h. Sampling plan A tests of MIL-L-25467 and MIL-L-27160

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three out of the first 15 and one item out of each additional 100 or less produced on contract or order shall be selected at random and subjected to the following tests as described under 4.5.2.1 and 4.6.

a. Sampling plan A tests . . . . . See 4.5.2.1

b. Vibration endurance . . . . . See 4.6.10

c. Shock . . . . . See 4.6.11

d. Mounting lug . . . . . See 4.6.12

e. High temperature exposure . . . . . See 4.6.13

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f. Low temperature exposure . . . . . See 4.6.14

g. Temperature shock . . . . . See 4.6.15

h. Humidity . . . . . See 4.6.16

i. Salt fog . . . . . See 4.6.17

j. Fungus resistance . . . . . See 4.6.18

k. Sampling plan B tests of MIL-L-25467 and MIL-L-27160

l. Internal examination . . . . . See 4.6.22

4.5.2.3 Sampling plan C. Unless otherwise specified (see 6.2), three accelerometers shall be selected at random from each 100 or less on the contract or order and shall be subjected to the following tests.

a. Individual tests . . . . . See 4.5.1

b. Production reliability test . . . . . See 4.7.2

4.5.3 Rejection. Items failing to conform to this specification shall be rejected.

4.6 Test methods

4.6.1 Examination of product. Each accelerometer shall be examined externally to determine conformance to this specification with respect to materials, markings, and workmanship (3.2, 3.7, and 3.13, respectively).

4.6.2 Static scale error and friction. The static scale error shall be observed with the accelerometer held in each of the following positions:

a. Plane of dial vertical with +5g graduation at top (scale reading shall be +1g).

b. Plane of dial horizontal (scale reading shall be zero g).

c. Plane of dial vertical with +5g graduation at bottom (scale reading shall be -1g).

In each position, the main pointer reading obtained shall be due entirely to gravity and not to motion in change of position of the accelerometer. The static scale error shall not exceed ±0.2g units at the a. (1g) and c. (-1g) test positions and ±0.3g units at the b. (0g) test position. Friction shall not exceed ±0.2g with the memory pointer engaged.

4.6.2.1 Dynamic scale error. The accelerometer shall be mounted on a suitable centrifuge and when the speeds are properly stabilized, determine the scale errors for accelerations of -5g, -3g, 0g, +3g, +6g, +9g, and at any other points desired by the procuring activity. The dynamic scale error shall not exceed ±0.2g at any specified test point.

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4.6.3 Lighting. The indicator shall be subjected to the individual lighting tests of MIL-L-25467 and MIL-L-27160.

4.6.3.1 Visual comparison. This test shall be conducted by qualified personnel only after they have been tested for normal color vision and have spent at least 20 minutes in the dark to become dark adapted. The prime standard and the indicator under test shall be placed side-by-side at eye level in completely dark surroundings and compared as described in the following paragraphs.

4.6.3.1.1 Brightness. With 2.50±0.01V applied to the indicator under test and 2.40 and 2.60V applied consecutively to the lighting terminals of the prime standard, like areas of the pointers, numerals, indicia, markings, et cetera, of the indicator under test and the prime standard shall be compared for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit at 2.50V shall fall within the brightness settings of the prime standard when set at 2.40 and 2.60V. If the brightness is dissimilar in any individual area or the lighting of the indicator and prime standard is not uniform, the questionable areas of the production units shall be measured with a photometer in accordance with MIL-L-25467 or MIL-L-27160, as applicable. If the production units fail this test, they shall be rejected and returned for rework. Preproduction units shall not be tested by use of the prime standard.

4.6.3.1.2 Color. The color comparison test shall be conducted as specified in 4.6.3.1.1, except 4.50V shall be applied to both the production indicator and the prime standard. The two indicators shall be identical in color.

4.6.4 Response characteristics. The accelerometer shall be subjected to vibration applied along the axis passing through the +5 and -3 graduations of the dial. The test apparatus used shall be as specified in 4.6.4.1. The accelerometer shall be tested at each frequency and amplitude listed in table II. During the test, total excursion of the pointer movement shall be recorded at each frequency stage of the test. The recorded value shall not exceed the maximum and minimum values listed in table II.

4.6.4.1 Test apparatus. The equipment shall impose linear sinusoidal motion to the accelerometer and shall be capable of producing the frequencies and amplitude specified in table II. The acceleration vector of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

4.6.5 Return mechanism operation. The accelerometer shall be placed in operating position with the pointer deflected to the maximum indication, the reset knob shall then be operated to return the pointer to zero. The force applied to the reset knob shall be between 3 to 8 pounds.

4.6.5.1 Locking mechanism operation. The accelerometer shall be locked with the auxiliary pointers approximately 3.0g from the main pointer, the readings noted, and the accelerometer shall then be subjected to the test specified in 4.6.11. The main pointer shall not have changed more than 0.2g. The locking mechanism shall then be unlocked.

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TABLE II. Response characteristic test settings and tolerances

Impressed frequency (Hz.)	Double Amplitude (inches)	*Indicated acceleration "g"	
		Minimum	Maximum
5	2.0	4.8	5.8
6	1.5	4.8	6.3
7	1.25	4.9	7.2
8	0.75	3.4	5.6
9	0.50	2.6	4.7
10	0.50	2.8	5.8
11	0.50	0	6.7
12	0.50	0	7.5
13	0.375	0	6.2
14	0.375	0	6.8
15	0.375	0	7.1
16	0.25	0	5.0
18	0.1875	0	3.9
20	0.1875	0	3.8

NOTE: Indicated acceleration is the positive acceleration reading plus the magnitude of the negative acceleration reading.

4.6.5.2 Force on knob. A force of 15 +1 pounds shall be applied perpendicular to the shaft at the extreme outward end of the setting knob. When the force is removed, the shaft and knob shall return to the original position without permanent deformation.

4.6.6 Low-temperature operation. The accelerometer shall be placed in a chamber and maintained at a temperature of -54° ±2°C for a period of 4 hours. Upon completion of the test period and while still at the test temperature, the accelerometer shall be subjected to and pass the test specified in 4.6.3.

4.6.7 High-temperature operation. The accelerometer shall be placed in a chamber and maintained at a temperature of 71° ±2°C for a period of 4 hours. Upon completion of the test period and while at the test temperature, the accelerometer shall be subjected to and pass the test specified in 4.6.3.

4.6.8 Magnetic effect. The accelerometer shall be held in various positions with the nearest part 5 inches from the magnetically East or West of the center of a short bar magnet compass. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The compass deflection for any position of the accelerometer shall not exceed 1 degree.

4.6.9 Vibration error. The accelerometer shall be mounted on a suitable vibration stand with the dial in a vertical plane and with the graduation +5 at the top. The auxiliary pointers shall be displaced approximately 2g from the main pointer and the readings noted. The reset mechanism shall not be operated until the completion of the test. For a total of 3 hours the accelerometer shall be vibrated at the specified frequencies at a double amplitude between

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0.018 and 0.020 inch, and the pointer oscillation shall be noted. The frequency shall be varied from 500 to 3,000 cpm in a period of not more than 60 minutes. The natural frequency of the seismic system shall be noted during the frequency variation between 500 to 3,000 cpm. For the remainder of the 3-hour test period, the accelerometer shall be vibrated at a frequency of 1,800 cpm. At the end of the test period, the indications of the auxiliary pointer shall be read before operating the resetting mechanism. At frequencies below 1,200 cpm, the oscillation of the main pointer shall not exceed 0.4g and at frequencies over 1,200 cpm, the oscillation shall not exceed 0.6g. After vibration, the indications of the auxiliary pointers shall not be changed by more than 0.2g. The accelerometer shall suffer no damage as a result of this test.

4.6.10 Vibration endurance. The accelerometer shall be mounted as specified for the vibration error test and the auxiliary pointers displaced approximately 2g from the main pointer. The indications of the auxiliary pointers should be noted and the reset mechanism shall not be operated until the completion of the test. The accelerometer shall be subjected to vibration at a double amplitude between 0.018 and 0.020 inch, in the frequency range of 500 to 3,000 cpm, preferably at the frequency at which maximum pointer oscillation is obtained, for a period of 50 hours. Oscillation of the main pointer shall not exceed 0.8g during or after the test. At the completion of the test, the indications of the auxiliary pointers shall not be changed by more than 0.4g. The accelerometers shall then be subjected to the test specified in 4.6.3. The dynamic scale error shall not exceed 0.2g.

4.6.11 Shock. The accelerometer shall be installed in the normal operating position in a free-fall apparatus as specified in 4.6.11.1. If provided with a locking device, the instrument mechanism shall be locked. The accelerometer shall be allowed to fall from a height of 8 inches, then examined for damage to the accelerometer, and the dynamic scale error shall not exceed 0.2g.

4.6.11.1 Apparatus. A rigidly constructed device provided with a panel mounting for installing the accelerometer in the specified position, and vertical guides to allow free-fall from the specified height shall be used. A bumper on the free-fall element shall arrest its fall by striking the center of a rounded projection on a metal block. The bumper shall be spring steel, 0.250 by 1.500 inches in cross section with an unsupported length of 5 inches between points of the midpoint of a broad side. The center of gravity of the accelerometer under test and the contact point of the bumper shall be aligned vertically within 0.125 inch. The weight of the free-fall element shall be approximately 6 pounds. The metal block shall weigh not less than 50 pounds.

4.6.12 Mounting lugs. The instrument case, with mechanism and cover glass removed, shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs received no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied along the axis of the mounting hole and toward the front of the case for a period of 1 minute. There shall be no fracture.

4.6.13 High-temperature exposure. The accelerometer shall be subjected to a temperature of 71° ±2°C for a period of 24 hours. After return to normal

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temperature, the accelerometer shall be subjected to th test specified in 4.6.3. There shall be no damage to the accelerometer, and the dynamic scale error shall not exceed 0.2g.

4.6.14 Low-temperature exposure. The accelerometer shall be maintained at a temperature of -65° ±2°C for a period of 48 hours. The temperature shall then be changed to -54° ±2°C and maintained for an additional 24-hour period. After return to normal temperature, the accelerometer shall be subjected to the test specified in 4.6.2. There shall be no damage to the accelerometer, and the static scale error shall not exceed 0.2g.

4.6.15 Temperature shock. The accelerometer shall be subjected to a temperature shock test in accordance with MIL-STD-810, Method 503.1, procedure I. At the completion of this test, the accelerometer shall be subjected to and shall meet the requirements specified in 4.5.1.

4.6.16 Humidity. The accelerometer shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.1, procedure I. At the completion of this test, the individual tests (4.5.1) shall be repeated. No damage shall result from this test and the tolerances specified for the individual tests shall apply.

4.6.17 Salt fog. The accelerometer shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.1, procedure I, except for a period of 50 hours. At the end of the 50-hour period, the individual tests (4.5.1) shall be repeated. No damage shall result from this test, and the tolerances specified for the individual tests shall apply.

4.6.18 Fungus. The accelerometer shall be subjected to a fungus test in accordance with MIL-STD-810, Method 508.1, procedure I, except for a period of 14 days. At the end of this test, the accelerometer shall be carefully examined to ascertain that no fungus growth has occurred that would affect subsequent operation.

4.6.19 Leakage (applicable to AFB-4B/A only). The accelerometer shall be tested for leaks using a mass spectrometer type of helium leak detector. The detected leak rate shall not exceed 0.01 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.6.20 Fogging (applicable to ABU-4B/A only). The accelerometer shall be energized and kept in a 70° ±2°C controlled ambient temperature for a minimum of 1 hour. At the end of this period, and while at the high temperature, an ice cube shall be rubbed on the face of the accelerometer glass for a period of 1 to 2 minutes. The glass shall then be wiped dry (compressed air shall not be used), and the accelerometer inspected for evidence of water and oil fog. Any evidence of fogging shall be cause for rejection.

4.6.21 Internal examination. The case of the accelerometer shall be opened and the mechanism shall be examined. Any deterioration or damage which could in any manner prevent the accelerometer from meeting functional operation requirements during service life shall be cause for rejection.

4.6.22 Lighting tests. The accelerometer shall be subjected to all the lighting tests specified in MIL-L-25467 and MIL-L-27160, as applicable.

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4.7 Reliability

4.7.1 Qualification reliability. The six samples specified in 4.4.3 shall be subjected to a reliability test in accordance with MIL-STD-781, test level F, test plan III, to demonstrate a minimum acceptable mean time between failures (MTBF) of 1000 hours. During the operative periods (periods A and B, figure 1 of MIL-STD-781) at approximately every 15 seconds, the accelerometers shall be subjected to an impact of sufficient magnitude to register alternately, at least, -3 and +5g units, with the auxiliary pointers reset by the PUSH TO SET knob every 2 minutes. Upon completion of each 150 operating hours the instrument shall be subjected to and shall meet the requirements of 4.6.2, 4.6.3, 4.6.4, and 4.6.5. A failure has occurred whenever a unit fails any test.

4.7.2 Production reliability tests. Samples of the accelerometer shall be tested as specified in 4.7.1, except that the accept-reject criteria for test plan V of MIL-STD-781 shall be applied.

4.7.3 Longevity test. Two accelerometers shall be tested in accordance with test plan XXVIII and test level F of MIL-STD-781, while cycling each accelerometer for 3000 hours in accordance with 4.7.1. At 200 hours operating time intervals, each accelerometer shall be subjected to and shall meet the requirements of 4.6.2, 4.6.3, 4.6.4, and 4.6.5. Test time accumulated on the equipment during a reliability test may be applied toward the longevity test when the same sample units are used.

4.8 Packaging inspection. The inspection of the preservation-packaging and packing shall be inspected to determine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Accelerometers (nonhermetically sealed) shall be packaged in accordance with MIL-P-116, Method IIC, one each in a unit container conforming to PPP-B-636, weather-resistant class with inside dimensions of 6 by 6 by 8 inches deep. Each unit container shall be lined with cushioning pads 2 inches thick, constructed of MIL-P-26514, class 2, 1.6 pound FT.3 density. Hermetically sealed units shall have water-vapor-proof caps applied to electrical connectors and shall be packaged in accordance with MIL-P-116, Method III. Cushioning and unit containers shall be identical to requirements specified for nonhermetically sealed units.

5.1.2 Level C. Accelerometers shall be provided sufficient protection against physical and mechanical damage to insure safe delivery, without degradation of the item reliability, from the supply source to the first domestic receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as applicable (see 6.2).

5.2.1 Level A. Accelerometers, packaged as specified in 5.1.1, shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class. Insofar as practicable, exterior containers shall be of uniform shape and size and of minimum cube and tare consistent with the protection required.

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5.2.2 Level B. Accelerometers, packaged as specified in 5.1.1, shall be packed in domestic class exterior containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 shall apply.

5.2.3 Level C. Level C packing shall afford adequate protection against damage during direct domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the suppliers commercial practice, provided the latter meets the requirements of this level.

5.3 Reinspection date marking. Reinspection marking shall be as specified by the procuring activity (see 6.2).

5.4 Precautionary marking. The following precautionary marking shall appear conspicuously on two opposite sides of each interior package and shipping container, wherever practicable:

"FRAGILE - DELICATE INSTRUMENTS
HANDLE WITH CARE".

6. NOTES

6.1 Intended use. The accelerometers covered by this specification are intended for use on aircraft to indicate vertical acceleration in maneuvers and in rough air.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements

a. Title, number, and date of this specification.

b. Type of accelerometer required (see 1.2)

c. Whether sampling plan B and C are to be omitted (see 4.5.2.2 and 4.5.2.3).

d. Where the preproduction inspection samples should be sent and activity responsible for testing (see 4.4.1).

e. Levels of packaging-packing desired (see 5.1 and 5.2).

f. Reinspection date marking, if required (see 5.3).

6.2.2 Contract data requirements. Data specified in 4.4.2 will be listed directly on a DD Form 1423 incorporated into the contract.

6.3 Preproduction test provisions. It is understood that the accelerometer supplied under contract or order will be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship, and method of manufacture. Deviation from the standards of the preproduction sample will be made only by the procuring activity. Evidence of unauthorized change will constitute cause for rejection.

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6.4 Definitions

6.4.1 Tapping. The phrase "tapped lightly" is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.4.2 Amplitude. Whenever the word "amplitude" is specified, it means the extent of motion as measured from mean to an extreme position.

6.4.3 "G" unit. The "g" is defined as an acceleration of 32.2 feet per second per second.

6.4.4 Scale error. Scale error is defined as the difference between the indication of the accelerometer and the corresponding calibrated standard.

6.5 International standardization agreement. Certain provisions (3.4.1, 3.4.5, 3.4.8, 3.4.9, and figure 1) of this specification are subject of international standardization agreement ASCC AIR STD 10/4A and STANAG 3330. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11
Navy - AS

Preparing activity:
Air Force - 11

Reviewer activity:
Air Force - 99

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International interest (see 6.5)

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